# Evidence-Gated Cross-Modal Invariance for Robust Vision-Language-Action Policies

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***Abstract*—Vision-Language-Action (VLA) models achieve strong performance via early multimodal fusion, but can overfit to spurious cross-modal correlations. We study state-dependent modality entanglement in $\pi_{0.5}$ on BEHAVIOR-1K and propose evidence-gated regularization that (i) enforces invariance to perturbations in non-evidence modalities and (ii) encourages single-modality sufficiency when evidence is strong. Evidence is derived from simulator segmentation and skill annotations, avoiding circular self-labeling.**

## I. Introduction

Vision-Language-Action (VLA) models have recently shown strong performance in robotic manipulation by jointly reasoning over multimodal inputs such as images, language, and proprioception. A key design choice in modern VLAs is early multimodal fusion, where tokens from different modalities are combined and processed jointly throughout the network. While this enables rich cross-modal interactions, it also introduces a fundamental challenge: *modality entanglement*, where the policy becomes overly dependent on correlations between modalities rather than true task-relevant signals. As a result, the learned policy may rely on spurious features and exhibit brittle behavior under distribution shift.

In this work, we identify two common failure modes of modality entanglement in VLAs. First, the policy can be highly sensitive to *useless modalities*: when a modality contains little or no task-relevant information, corrupting or removing it can still significantly degrade performance. This indicates that the model has overfit to spurious cross-modal correlations rather than learning to ignore irrelevant inputs. Second, even when multiple modalities are informative, the policy may fail when one of them is corrupted, despite the remaining modalities containing sufficient information for the task. This reflects a lack of *single-modality sufficiency*, where the policy over-relies on joint multimodal co-occurrence and cannot fall back to a single reliable modality.

Both failure modes directly impact the robustness and generalization of VLA policies, especially in real-world settings where sensors may be noisy, occluded, or partially unavailable. To address these issues, we propose an evidence-gated training framework that explicitly regularizes modality usage. Our approach introduces two complementary objectives: (1) an invariance loss that suppresses sensitivity to modalities with low task-relevant evidence, and (2) a sufficiency loss that encourages the policy to succeed when only a single high-evidence modality is available. By leveraging simulator-derived evidence signals, our method provides a principled way to reduce harmful modality entanglement while preserving useful redundancy.

## II. Preliminary Diagnosis and Benchmark Design

We use two diagnostics.

### A. Inference-time Missing-Modality Sensitivity

Given a fixed observation, we compare policy outputs under: (i) all modalities, (ii) single-modality removal (global/left/right), and (iii) joint wrist removal. We measure action deviation and segment success. We observe that global view is frequently critical for navigation, manipulation often collapses when removing the active-arm camera, and sometimes removing a seemingly irrelevant modality *improves* success, indicating non-monotonic interactions.

### B. Segment Rollout with Fixed Occlusion

We evaluate paired rollouts within predefined segments using persistent rectangular occlusion (fixed mask for the attacked modality across the segment). Table I shows representative results confirming systematic patterns: global view is essential for navigation; wrist views can both help and harm; manipulation is strongly sensitive to removing a single arm camera.

### C. Benchmark Suite (Desired Final Benchmark)

Building on the preliminary diagnostics above, we aim to standardize evaluation into three complementary benchmark sets:

- **Set 1: Inference-only evaluation on demonstrations.** Evaluate missing-modality and modality-corruption sensitivity on demonstration observations (no closed-loop rollout), measuring action/flow deviation and per-segment success proxies.
- **Set 2: Random erasing + rollout evaluation.** Apply stochastic modality degradation (e.g., random erasing / noise / blackout) during closed-loop rollouts and report paired segment-level success, focusing on robustness to partial sensor corruption.
- **Set 3: Real occlusion tasks (train clean, deploy occluded).** Evaluate scenarios where, at deployment, the task-relevant object is occluded in a particular view (e.g., head/global camera) while remaining views (e.g., wrist cameras) still observe it. The goal is to diagnose modality sufficiency failure: the policy should succeed by relying on the non-occluded modalities even though training data contains no explicit occlusion augmentation.

Across all sets we use paired seeds and segment definitions (navigation vs manipulation) to isolate state-dependent modality entanglement and robustness gaps.

TABLE I

SEGMENT-LEVEL ROLLOUT SUCCESS RATES UNDER FIXED MODALITY OCCLUSION (20 RUNS PER SEGMENT). WE REPORT NAVIGATION (NAV) AND MANIPULATION (MAN) SEGMENTS SEPARATELY.

| **MAN (7 segments)** | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| Task | Inst | Block | Seg | Checker | Full | No G | No L | No R | No W |
| radio | inst_211 | B001 | 750:1250 | picked_up[radio_receiver] | 30% | 5% | 0% | 0% | 0% |
| radio | inst_211 | B002 | 1250:1750 | radio_toggled_on[radio_receiver] | 0% | 0% | 0% | 0% | 0% |
| trash | inst_196 | B001 | 2750:3250 | picked_up[ashcan] | 40% | 5% | 25% | 0% | 0% |
| trash | inst_196 | B003 | 4000:4750 | picked_up[can_soda] | 70% | 5% | 0% | 55% | 10% |
| trash | inst_196 | B004 | 5000:6000 | picked_up[can_soda] | 55% | 35% | 0% | 70% | 0% |
| trash | inst_196 | B005 | 6250:8000 | picked_up[can_soda] | 70% | 80% | 0% | 95% | 0% |
| trash | inst_67 | B002 | 1250:1750 | picked_up[ashcan] | 25% | 5% | 10% | 0% | 0% |
| Avg | | | | | 41.4% | 19.3% | 5.0% | 31.4% | 1.4% |
| **NAV (4 segments)** | | | | | | | | | |
| trash | inst_196 | B000 | 250:2750 | reach[ashcan] | 85% | 25% | 95% | 100% | 65% |
| trash | inst_196 | B002 | 3250:4000 | reach[can_soda] | 100% | 0% | 15% | 80% | 0% |
| trash | inst_67 | B000 | 250:1250 | reach[ashcan] | 75% | 0% | 95% | 100% | 10% |
| trash | inst_67 | B001 | 1750:2750 | reach[can_soda] | 95% | 5% | 20% | 85% | 5% |
| Avg | | | | | 88.8% | 7.5% | 56.2% | 91.2% | 20.0% |

## III. METHODOLOGY

### A. Baselines

We compare:

1) $\pi_{0.5}$ official checkpoint
2) $\pi_{0.5}$ + **structural dropout** (removes modality tokens from attention graph)
3) $\pi_{0.5}$ + **content masking** (replaces missing modality tokens with learned modality-specific embeddings, preserving the attention graph)

### B. Frame-level Evidence from Simulator Annotations

For each timestep $t$ and camera modality $m \in \mathcal{M}$ (head/left/right), the dataset provides instance segmentation and skill annotations.

For each skill segment we define a set of *focal* task-relevant objects. For manipulation skills this corresponds to the annotated manipulated object(s). For navigation skills (e.g., `move to`), where no manipulated object exists, the navigation target object(s) are treated as the focal objects. In the notation below we keep the symbol $A^{man}$ to denote the visible area ratio of these focal objects (not necessarily physically manipulated). Let

$$A^{man}_{t,m,k} \in [0,1] \quad \text{area ratio of manipulated object } k, \tag{1}$$

$$A^{other}_{t,m,k} \in [0,1] \quad \text{area ratio of other annotated object } k. \tag{2}$$

Aggregate per-frame area ratios:

$$a^{man}_{t,m} = \sum_k A^{man}_{t,m,k}, \tag{3}$$

$$a^{other}_{t,m} = \sum_k A^{other}_{t,m,k}. \tag{4}$$

**Interaction gate.** To avoid large receptacles (e.g., table) dominating evidence when the manipulated object is absent, we only count "other" objects when the manipulated object is visible in that camera:

$$g_{t,m} = \mathbf{1}\big[a^{man}_{t,m} > \epsilon_{man}\big]. \tag{5}$$

Define the per-camera interaction score:

$$s_{t,m} = a^{man}_{t,m} + \alpha\, g_{t,m}\, a^{other}_{t,m}, \tag{6}$$

Let $s^{max}_t = \max_m s_{t,m}$. Relative evidence (normalized across cameras):

$$e_{t,m} = \frac{s_{t,m}}{\max_{m'} s_{t,m'} + \varepsilon} \in [0,1]. \tag{7}$$

We also define a global evidence gate to avoid applying auxiliary losses during "search" frames where no modality contains relevant evidence:

$$G_t = \mathbf{1}\big[s^{max}_t > \epsilon_{global}\big]. \tag{8}$$

### C. Proposed Objective

Let $o_t$ be the multimodal observation at time $t$, and $v_\theta(o_t)$ be the predicted flow (velocity) used by the $\pi_{0.5}$ flow-matching objective. Let $\tilde{o}^{(m)}_t$ denote $o_t$ with modality $m$ perturbed (appearance corruption / domain randomization), and $o^{(\text{keep } m)}_t$ denote the observation where *only* modality $m$ is kept intact while *all other* modalities are perturbed (or replaced by missing tokens).

We use two evidence thresholds $\tau_{\text{low}} < \tau_{\text{high}}$ and define hinge weights:

$$w_{\text{inv}}(e_{t,m}) = \max\left(0, \frac{\tau_{\text{low}} - e_{t,m}}{\tau_{\text{low}}}\right), \tag{9}$$

$$w_{\text{solo}}(e_{t,m}) = \max\left(0, \frac{e_{t,m} - \tau_{\text{high}}}{1 - \tau_{\text{high}}}\right). \tag{10}$$

*a) (1) Evidence-gated nuisance invariance.:* When evidence for modality $m$ is low, enforce invariance to perturbations in $m$:

$$\mathcal{L}_{inv} = \sum_t \sum_{m \in \mathcal{M}} G_t\, w_{\text{inv}}(e_{t,m})\, \big\| v_\theta(o_t) - v_\theta(\tilde{o}^{(m)}_t) \big\|_2^2. \tag{11}$$

*b) (2) Single-modality sufficiency (solo).:* When evidence for modality $m$ is strong, enforce consistency when all other modalities are degraded:

$$\mathcal{L}_{solo} = \sum_t \sum_{m \in \mathcal{M}} G_t \, w_{\mathrm{solo}}(e_{t,m}) \left\| v_\theta(o_t) - v_\theta(o_t^{(\mathrm{keep}\, m)}) \right\|_2^2. \tag{12}$$

We choose a high $\tau_{\mathrm{high}}$ and a small weight on $\mathcal{L}_{solo}$ to avoid over-regularization.

*c) Final objective.:*

$$\mathcal{L} = \mathcal{L}_{flow} + \lambda_{inv}\mathcal{L}_{inv} + \lambda_{solo}\mathcal{L}_{solo}. \tag{13}$$

## IV. Benchmark Set 3 – Viewpoint-Consistent Occlusion

We construct a benchmark that tests whether a VLA can recover from head-camera occlusion by leveraging redundant wrist-camera observations. For each manipulation segment in the BEHAVIOR-1K demonstration set, we search for distractor object placements that substantially occlude the head camera's view of the manipulation target while leaving both wrist cameras unoccluded. The pipeline has three phases: segment identification, placement search, and benchmark compilation.

### A. Segment Identification

A *segment* is a manipulation skill instance identified by the tuple (task_id, skill_description, object_name, occurrence_idx), where occurrence index disambiguates repeated (description, object) pairs within an episode. We match segments semantically rather than by absolute skill index, because episodes of the same task can have variable skill counts due to inserted navigation primitives.

From the 12 target tasks ($\sim$75 qualified segments total), we retain segments that satisfy:

1) At least 20 usable episodes (excluding episodes with truncated or missing HDF5 data).
2) Baseline head-camera visibility $\geq 0.5\%$ area ratio (the target object is visible in the head camera at the skill start frame, measured from precomputed segmentation statistics).

### B. Placement Search

For each qualified segment, we search for distractor placements via the procedure in Algorithm 1. A distractor object is sampled from a curated pool of BEHAVIOR-1K assets and inserted into the simulator as a visual-only USD primitive (no physics interactions), then evaluated for selective occlusion.

*a) Pose sampling.:* Candidate distractor positions are sampled along the ray from the head camera $\mathbf{c}_{\mathrm{head}}$ to the target object center $\mathbf{p}_{\mathrm{target}}$:

$$\mathbf{p}_{\mathrm{dist}} = \mathbf{c}_{\mathrm{head}} + t \cdot (\mathbf{p}_{\mathrm{target}} - \mathbf{c}_{\mathrm{head}}) + \boldsymbol{\epsilon}, \tag{14}$$

where $t \sim \mathcal{U}(0.3, 0.7)$ controls the parametric distance along the ray and $\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathrm{diag}(\sigma_x^2, \sigma_y^2, \sigma_z^2))$ adds lateral/vertical perturbation ($\sigma_{\mathrm{lateral}}$=0.05 m, $\sigma_{\mathrm{vertical}}$=0.03 m). Yaw orientation is sampled uniformly in $[0, 2\pi)$. Candidates whose axis-aligned bounding box (AABB) intersects the target's AABB are rejected to prevent physical overlap.

**Algorithm 1** Distractor Placement Search (per segment)

**Require:** Segment $s$ with usable episodes $\mathcal{E}$, distractor spec $d$
**Require:** Thresholds $\theta_{\mathrm{head}}$, $\theta_{\mathrm{wrist}}$; targets $N_{\mathrm{good}}$, $N_{\mathrm{pose}}$, $N_{\mathrm{ep}}$
**Ensure:** Set of accepted placements $\mathcal{P}$

$\mathcal{P} \leftarrow \emptyset$
**for** $i = 1$ **to** $N_{\mathrm{pose}}$ **do** ▷ Outer: pose variations
  Sample pose $(\mathbf{p}, \mathbf{q})$ via Eq. (14)
  **for** each $e \in$ SAMPLE$(\mathcal{E}, N_{\mathrm{ep}})$ **do** ▷ Inner: episodes
    RESTORESTATE($e$, $s$.start_frame)
    $\{I_k^{\mathrm{bl}}\} \leftarrow$ RENDERALLCAMERAS
    ADDDISTRACTOR($d$, $\mathbf{p}$, $\mathbf{q}$) ▷ Visual-only USD prim
    $\{I_k^{\mathrm{dt}}\} \leftarrow$ RENDERALLCAMERAS
    $\{o_k\} \leftarrow$ COMPUTEOCCLUSION($\{I_k^{\mathrm{bl}}\}$, $\{I_k^{\mathrm{dt}}\}$, $s$.AABB)
    **if** $o_{\mathrm{head}} \geq \theta_{\mathrm{head}}$ **and** $\forall w$: $o_w \leq \theta_{\mathrm{wrist}}$ **then**
      $\mathcal{P} \leftarrow \mathcal{P} \cup \{(e, \mathbf{p}, \mathbf{q}, \{o_k\})\}$
    **end if**
    REMOVEDISTRACTOR
    **if** $|\mathcal{P}| \geq N_{\mathrm{good}}$ **then break outer**
    **end if**
  **end for**
**end for**
**return** $\mathcal{P}$

*b) Occlusion measurement.:* Rather than relying on instance segmentation (which is unavailable in headless rendering mode), we compute occlusion via RGB differencing. For each camera $k \in \{\mathrm{head}, \mathrm{left\_wrist}, \mathrm{right\_wrist}\}$:

1) Project the target's 3D AABB onto the camera's image plane to obtain a 2D bounding box $B_k$.
2) Render baseline RGB $I_k^{\mathrm{bl}}$ (no distractor) and distractor RGB $I_k^{\mathrm{dt}}$.
3) Compute per-pixel mean absolute difference within $B_k$; pixels exceeding a threshold $\tau$=25 are counted as occluded.
4) Occlusion ratio: $o_k = n_{\mathrm{changed}}/n_{\mathrm{total}}$ within $B_k$.

A placement is *accepted* if $o_{\mathrm{head}} \geq \theta_{\mathrm{head}}$ and $o_{\mathrm{left\_wrist}} \leq \theta_{\mathrm{wrist}}$ and $o_{\mathrm{right\_wrist}} \leq \theta_{\mathrm{wrist}}$, with default thresholds $\theta_{\mathrm{head}}$=0.6 and $\theta_{\mathrm{wrist}}$=0.1.

*c) Search strategy.:* The outer loop iterates over pose variations; the inner loop tests each pose across multiple episodes. This *outer=poses, inner=episodes* ordering finds placements that generalize across different initial states rather than episode-specific solutions. Each segment has a budget of $N_{\mathrm{pose}} \times N_{\mathrm{ep}}$ attempts (default $20 \times 20 = 400$); search terminates early once $N_{\mathrm{good}}$ (default 10) accepted placements are found.

### C. Benchmark Compilation

Per-segment placement results are aggregated into a single benchmark configuration. Segments with fewer than $N_{\mathrm{min}}$ (default 5) accepted placements are excluded. Each accepted placement records the distractor's asset identity, SE(3) pose, and per-camera occlusion ratios. No additional simulator state files are saved; at evaluation time, the original HDF5 state is restored and the distractor is spawned at the recorded pose.

TABLE II
HYPERPARAMETERS FOR EVIDENCE EXTRACTION AND AUXILIARY LOSSES.

| Hyperparam | Meaning | Tuning trend / guidance |
|---|---|---|
| $\alpha$ | Weight on "other" (context) objects when the focal object is visible | Increase if context objects help disambiguate once the focal object is in view; decrease if large receptacles dominate evidence. |
| $\epsilon_{man}$ | Visibility threshold for the focal object to activate $g_{t,m}$ | Increase to be stricter; decrease if focal objects are small and often undercounted. |
| $\epsilon_{global}$ | Global evidence threshold on $s_t^{max}$ for $G_t$ | Increase to skip more "search/no-evidence" frames; decrease if you skip too many informative frames (monitor fraction of $G_t = 1$). |
| $\tau_{\text{low}}$ | Low-evidence cutoff for invariance hinge $w_{\text{inv}}$ | Smaller: invariance only near zero evidence; larger: invariance applies more broadly (risk over-regularization). |
| $\tau_{\text{high}}$ | High-evidence cutoff for solo hinge $w_{\text{solo}}$ | Larger: solo more conservative; smaller: solo enforced more often (risk if evidence is noisy). |
| $\lambda_{inv}$ | Weight of $\mathcal{L}_{inv}$ | Increase to reduce nuisance sensitivity; if overall performance drops, reduce or lower $\tau_{\text{low}}$. |
| $\lambda_{solo}$ | Weight of $\mathcal{L}_{solo}$ | Keep small; increase only if sufficiency failure persists when a modality is clearly evidential. |
| $\varepsilon$ | Numerical stabilizer in normalization | Fixed small constant; change only if encountering numerical issues. |
| Corruption type/strength | How $\tilde{o}_t^{(m)}$ and non-kept modalities are degraded | Start simple (blackout / random erasing) with moderate severity; increase to match evaluation shifts, but avoid unrealistic inputs. |

## V. Project Plan (March–April 2026)

As of March 24, 2026, approximately one month remains before the end of the internship (April 27, 2026).

### A. Simulation Experiments (Ongoing)

- Complete evaluation of current training runs (including converged checkpoints).
- Verify whether the proposed method yields **consistent performance improvements** across:
  - Missing-modality settings (evaluating $\mathcal{L}_{solo}$)
  - Nuisance-modality corruption settings (evaluating $\mathcal{L}_{inv}$)
- Current results (from partially converged checkpoints) already show promising trends; we will confirm these trends with fully converged models.
- If consistent improvements are observed, scale evaluation to a **larger set of tasks** within BEHAVIOR-1K to strengthen statistical validity.

### B. Real-Robot Experiments (Primary Focus)

To directly validate the two motivations, we design two types of real-robot experiments aligned with $\mathcal{L}_{inv}$ and $\mathcal{L}_{solo}$ respectively.

*1) Nuisance-Modality Robustness ($\mathcal{L}_{inv}$) via Multi-View Vision:* We leverage an existing multi-view setup (third-person + wrist cameras).

**Task.** We simplify a long-horizon furniture assembly task into a **single skill**: placing a rear component onto a chair. This task primarily relies on the right arm and third-person view.

**Setup.**

- The baseline VLA is known to overfit to all camera inputs.
- We introduce **nuisance corruption** on an irrelevant modality (e.g., left wrist camera) by placing out-of-distribution visual distractors.

**Hypothesis.**

- Baseline policy fails due to spurious dependence on the corrupted modality.
- Our method maintains performance by learning to ignore nuisance modalities.

*2) Single-Modality Sufficiency ($\mathcal{L}_{solo}$) via Vision + Tactile:* We design a contact-rich task combining vision and tactile sensing.

**Task.** The robot grasps a cable and moves along it to locate a target position, then lifts the cable. The cable is:

- mostly smooth,
- with a **distinct tactile feature** near the target location.

**Data collection.**

- The initial gripper pose is randomized along the cable.
- Each demonstration consists of:
  - closing the gripper,
  - moving along a fixed direction,
  - stopping at the correct tactile location,
  - lifting the cable.
- Both vision and tactile modalities are used during training.

**Deployment.**

- Visual input is degraded (e.g., lights turned off).
- Optionally include visual cues (e.g., color markers) that the baseline may overfit to.

**Hypothesis.**

- Baseline policy fails due to reliance on visual signals.
- Our method succeeds by **falling back to tactile sensing**.

### C. Timeline Summary

- **Late March:** finalize simulation evaluation and scaling.
- **Early April:** implement and validate $\mathcal{L}_{inv}$ real-robot experiment.
- **Mid April:** implement $\mathcal{L}_{solo}$ tactile experiment (if setup is stable).
- **Late April:** finalize analysis, ablations, and paper writing.